ON SIGN PATTERNS OF FOURIER COEFFICIENTS OF THETA QUOTIENTS

KOUSTAV BANERJEE, KATHRIN BRINGMANN, BERNHARD HEIM, AND BEN KANE

Abstract. Motivated by conjectures of Schlosser and Zhou on sign patterns of Fourier coefficients of infinite products, we study two families of theta quotients and their positive integral powers. We determine asymptotic formulas for their Fourier coefficients and use them to obtain explicit eventual periodic sign patterns. Our results provide new instances of the sign phenomena observed by Schlosser and Zhou and illustrate how dominant cusps determine the resulting sign patterns.

1. Introduction and statement of results

In this paper, for $m \in \mathbb{N}_{\geq 3}$ and $\ell \in \mathbb{Z} \setminus \{0\}$, we study the sign pattern of the Fourier coefficients for the quotient

$$\frac{\left(q, q^{m-1}; q^m\right)_\infty^\ell}{\left(q^2, q^{m-2}; q^m\right)_\infty^\ell}. \tag{1.1}$$

Here and throughout, $|q| < 1$ and for $a \in \mathbb{C}$ and $n \in \mathbb{N}_0 \cup \{\infty\}$, we set $(a)_n = (a;q)_n := \prod_{j=0}^{n-1}(1-aq^j)$. For $r \in \mathbb{N}$ and $a_1, \dots, a_r \in \mathbb{C}$, we let $(a_1, \dots, a_r)_n := (a_1, \dots, a_r; q)_n := (a_1)_n \dots (a_r)_n$. The primary motivation for considering this problem goes back to the sign pattern of Fourier coefficients of the Rogers–Ramanujan continued fraction

$$R(q) := \cfrac{q^{\frac{1}{5}}}{1 + \cfrac{q}{1 + \cfrac{q^2}{1 + \cfrac{q^3}{1 + \cdots}}}} = q^{\frac{1}{5}} \frac{(q, q^4; q^5)_\infty}{(q^2, q^3; q^5)_\infty}.$$

In this case, using the Circle method, the sign pattern of the n-th Fourier coefficient for n sufficiently large was proved by Richmond and Szekeres [16, (3.9)]. Andrews [2, Corollary 1] then gave a proof for the sign pattern of all coefficients via q-series dissections. Later, using dissections of Ramanujan's theta functions, Ramanathan [15, Theorem 5 and Theorem 6] and Chan–Yesilyurt [7, Theorem 1.2 and Theorem 1.3] studied periodicity of the sign of Fourier coefficients for a family of infinite products resembling (1.1). Recently, using similar techniques, Baruah and Sarma [3, Theorem 2 and Theorem 3] investigated the sign pattern of the coefficients of (1.1) with $m = 5$ and $\ell = \pm 5$. Using the Circle method, Schlosser and Zhou [17] studied the sign pattern of Fourier coefficients of theta quotients similar to (1.1) and made several conjectures. Three of the authors [4, Theorem 1.2] settled one of the conjectures of Schlosser and Zhou [17, Conjecture 22]. Other results in this direction can be found in [8, Section 2] and [11].

Returning to (1.1), for $m \in \mathbb{N}_{\geq 3}$ and $\ell \in \mathbb{N}$ we write

$$F_m^\ell(q) := \frac{\left(q, q^{m-1}; q^m\right)_\infty^\ell}{\left(q^2, q^{m-2}; q^m\right)_\infty^\ell} =: \sum_{n \geq 0} c_m^{[\ell]}(n) q^n \tag{1.2}$$

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and study the asymptotic growth and sign pattern of $c_m^{[\ell]}(n)$ as $n \to \infty$.

We first consider the case m odd. For $m = 3$, we trivially have $F_m^{\ell}(q) = 1$ and hence $c_m^{[\ell]}(n) = 0$ for all $n \in \mathbb{N}$. For $m \in \mathbb{N}_{>3}$ odd, define

$$\phi_m := \frac{1}{2}\left(1 - \frac{2}{m}\right)\left(1 - \frac{3}{m}\right). \tag{1.3}$$

Our first result is the following:

Theorem 1.1. *Let $m \in \mathbb{N}_{>3}$ be odd and $\ell \in \mathbb{N}$. Then, as $n \to \infty$,*

$$c_m^{[\ell]}(n) \sim \left(\frac{\left(1 - \frac{1}{m}\right)\left(1 - \frac{3}{m}\right)\ell}{2m}\right)^{\frac{1}{4}} \frac{(-1)^n \cos\left(\pi\left(\phi_m \ell - \frac{n}{m}\right)\right) e^{\pi\sqrt{\frac{2\left(1-\frac{1}{m}\right)\left(1-\frac{3}{m}\right)\ell}{m}n}}}{n^{\frac{3}{4}}}.$$

In particular, for $n \gg 1$,

$$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right) = (-1)^n \operatorname{sgn}\left(\cos\left(\pi\left(\phi_m \ell - \frac{n}{m}\right)\right)\right).$$

Next, we explicitly determine the sign in Theorem 1.1. For a property P, define

$$\chi_P := \begin{cases} 1 & \text{if } P \text{ holds,} \\ 0 & \text{otherwise.} \end{cases}$$

Moreover let

$$\mathcal{S}_{m,1}^{[\ell]} := \begin{cases} \left[\frac{m+1}{2} - \frac{5\ell}{2}, \frac{3m-1}{2} - \frac{5\ell}{2}\right] & \text{if } \ell \text{ is even,} \\ \left[m - \frac{5\ell-1}{2}, 2m - \frac{5\ell+1}{2}\right] & \text{if } \ell \text{ is odd.} \end{cases}$$

Corollary 1.2. *Let $\ell \in \mathbb{N}$, $m \in \mathbb{N}_{>3}$ odd with $m > 6\ell$, and define $0 \le r \le 2m-1$ via $n \equiv r \pmod{2m}$. Then, for $n \gg 1$,*

$$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right) = (-1)^{r + \left\lfloor \frac{\ell}{2} \right\rfloor + \chi_{r \in \mathcal{S}_{m,1}^{[\ell]}}}.$$

Next, we consider the growth of $c_m^{[\ell]}(n)$ for $m \in \mathbb{N}_{\ge 4}$ even and $\ell \in \mathbb{N}$.

Theorem 1.3. *We have, for $m \in \mathbb{N}_{\ge 4}$ even and $\ell \in \mathbb{N}$, as $n \to \infty$,*

$$c_m^{[\ell]}(n) \sim \frac{\ell^{\frac{1}{4}}}{2^{\ell + \frac{5}{4}} m^{\frac{1}{4}} \sin^{\ell}\left(\frac{2\pi}{m}\right)} \frac{(-1)^n e^{\pi\sqrt{\frac{2\ell}{m}n}}}{n^{\frac{3}{4}}}.$$

In this case, we have the following sign-pattern of $c_m^{[\ell]}(n)$.

Corollary 1.4. *We have, for $m \in \mathbb{N}_{\ge 4}$ even, $\ell \in \mathbb{N}$, and $n \in \mathbb{N}_0$,*

$$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right) = (-1)^n.$$

We next turn to $\ell < 0$. Writing $\ell = -\lambda$ with $\lambda \in \mathbb{N}$, for $m \in \mathbb{N}_{>3}$ with $\gcd(m, 6) = 1$, set

$$\mathcal{F}_m^{\lambda}(q) := \frac{\left(q^2, q^{m-2}; q^m\right)_{\infty}^{\lambda}}{\left(q, q^{m-1}; q^m\right)_{\infty}^{\lambda}} =: \sum_{n \ge 0} \gamma_m^{[\lambda]}(n) q^n.$$

If $m \equiv \pm 1 \pmod 6$ and $\lambda \in \mathbb{N}$, define

$$T_m^{[\lambda]}(n) := \cos\left(\pi\left(\frac{\lambda}{6} - \frac{n}{3} \pm \left(\frac{2n - \lambda}{6m} + \frac{\lambda}{m^2}\right)\right)\right).$$

We show the following result:

Theorem 1.5. *Let $m \in \mathbb{N}_{>3}$ with $\gcd(m,6)=1$ and $\lambda \in \mathbb{N}$. Then, as $n \to \infty$,*

$$\gamma_m^{[\lambda]}(n) \sim \left(\frac{\left(1-\frac{1}{m}\right)\left(1+\frac{1}{m}\right)\lambda}{6m}\right)^{\frac{1}{4}} \frac{T_m^{[\lambda]}(n) e^{\pi\sqrt{\frac{2\left(1-\frac{1}{m}\right)\left(1+\frac{1}{m}\right)\lambda}{3m}n}}}{n^{\frac{3}{4}}}.$$

To state the sign pattern of $\gamma_m^{[\lambda]}(n)$ for $m \equiv \pm 1 \pmod 6$, we define

$$\mathcal{S}_{m,2}^{[\lambda],\pm} := \begin{cases} \left[\frac{m+1}{2}+\frac{\lambda}{6}, \frac{3m-1}{2}+\frac{\lambda}{6}\right] & \text{if } \lambda \equiv 0 \pmod 6, \\ \left[\frac{(3\mp 1)m+4}{6}+\frac{\lambda-1}{6}, \frac{(9\mp 1)m-2}{6}+\frac{\lambda-1}{6}\right] & \text{if } \lambda \equiv 1 \pmod 6, \\ \left[\frac{(3\mp 2)m+5}{6}+\frac{\lambda-2}{6}, \frac{(9\mp 2)m-1}{6}+\frac{\lambda-2}{6}\right] & \text{if } \lambda \equiv 2 \pmod 6, \\ \left[\frac{\lambda+3}{6}, m+\frac{\lambda-3}{6}\right] & \text{if } \lambda \equiv 3 \pmod 6, \\ \left[\frac{(3\pm 2)m+7}{6}+\frac{\lambda-4}{6}, \frac{(9\pm 2)m+1}{6}+\frac{\lambda-4}{6}\right] & \text{if } \lambda \equiv 4 \pmod 6, \\ \left[\frac{(3\pm 1)m+8}{6}+\frac{\lambda-5}{6}, \frac{(9\pm 1)m+2}{6}+\frac{\lambda-5}{6}\right] & \text{if } \lambda \equiv 5 \pmod 6. \end{cases}$$

Corollary 1.6. *Let $\lambda \in \mathbb{N}$, $m \in \mathbb{N}_{>3}$ with $m \equiv \pm 1 \pmod 6$ and $\lambda < \frac{m}{2}$, and assume that $n \equiv 3r + 2mj \pmod{6m}$ with $0 \le r \le 2m-1$ and $0 \le j \le 2$. Then, for $n \gg 1$,*

$$\operatorname{sgn}\left(\gamma_m^{[\lambda]}(n)\right) = (-1)^{r+\left\lfloor \frac{\lambda}{6}\right\rfloor + \chi_{r \in \mathcal{S}_{m,2}^{[\lambda],\pm}}} \begin{cases} 1 & \text{if } \lambda \equiv 0,1,2 \pmod 6, \\ -1 & \text{if } \lambda \equiv 4,5 \pmod 6, \\ \left(\frac{m}{3}\right) & \text{if } \lambda \equiv 3 \pmod 6, \end{cases}$$

where $\left(\frac{\cdot}{\cdot}\right)$ denotes the Jacobi-symbol.

The paper is organized as follows. In Section 2, we state an exact formula of Zuckerman, preliminary results on the Jacobi theta function, and Ingham's Tauberian theorem. Proofs of Theorem 1.1 and Corollary 1.2 are given in Section 3. We show Theorem 1.3 and Corollary 1.4 in Section 4. Section 5 is devoted to the proof of Theorem 1.5 and Corollary 1.6. Finally, we conclude this paper with Section 6 by proposing questions for future research.

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2. Preliminaries

2.1. **Exact formulas of Rademacher and Zuckerman.** We require an exact formula for Fourier coefficients of weakly holomorphic modular functions. Rademacher [13] perfected the Circle Method of Hardy and Ramanujan [10] and obtained an exact formula for the partition function. This was later extended to weakly holomorphic forms of nonpositive weight in results of Rademacher–Zuckerman [14] and Zuckerman [18] and by Ono and the second author [6]. To state the result, let I_α denote the usual I-Bessel function. We allow weakly holomorphic modular forms with a multiplier system. Throughout, for $a \in \mathbb{N}$, $b \in \mathbb{Z}$, with $\gcd(a,b)=1$, we denote by $[b]_a$ the inverse of $b \pmod a$.

Theorem 2.1. *Let $\kappa \le 0$. Suppose that f is a weakly holomorphic modular form of weight κ on a subgroup Γ of finite index in $\mathrm{SL}_2(\mathbb{Z})$. Assume that, for some $\delta \ge 0$, $f(\tau) = \sum_{n \gg -\infty} a(n) q^{n+\delta}$. For*

$h, k \in \mathbb{N}_0$ *with* $0 \le h < k$ *and* $\gcd(h,k) = 1$*, define* $f_{h,k}$ *by*

$$f\left(\frac{1}{k}\left(h + iz\right)\right) = \left(\frac{i}{z}\right)^{\kappa} f_{h,k}\left(\frac{1}{k}\left(\left[-h\right]_k + \frac{i}{z}\right)\right) \quad (\mathrm{Re}(z) > 0)\,.$$

Write $f_{h,k}(\tau) =: \sum_{r \gg -\infty} a_{h,k}(r) q^{\frac{r+\delta_{h,k}}{c_k}}$*. Here* $c_k \in \mathbb{N}$ *and* $\delta_{h,k} \in [0,1)$*. Then, we have, for* $n + \delta > 0$*,*

$$a(n) = 2\pi i^{\kappa} (n+\delta)^{\frac{\kappa-1}{2}} \sum_{k \ge 1} \frac{1}{k} \sum_{\substack{0 \le h < k \\ \gcd(h,k)=1}} e^{-\frac{2\pi i (n+\delta) h}{k}}$$

$$\times \sum_{r + \delta_{h,k} < 0} a_{h,k}(r) e^{\frac{2\pi i}{k c_k}(r+\delta_{h,k})[-h]_k} \left(\frac{|r+\delta_{h,k}|}{c_k}\right)^{\frac{1-\kappa}{2}} I_{-\kappa+1}\left(\frac{4\pi}{k}\sqrt{\frac{|r+\delta_{h,k}|(n+\delta)}{c_k}}\right).$$

2.2. **The Jacobi theta function.** Recall the *Jacobi theta function*

$$\vartheta(w;\tau) := \sum_{n \in \mathbb{Z} + \frac{1}{2}} q^{\frac{n^2}{2}} e^{2\pi i n\left(w + \frac{1}{2}\right)} \quad \left(w \in \mathbb{C},\ q := e^{2\pi i \tau}\right).$$

Let $z \in \mathbb{C}$ with $\mathrm{Re}(z) > 0$, $k \in \mathbb{N}$, $h \in \mathbb{Z}$ with $\gcd(h,k) = 1$. Define $\omega_{h,k}$ by[1]

$$\eta\left(\frac{1}{k}(h+iz)\right) = e^{\frac{\pi i}{12k}(h - [-h]_k)} \omega_{h,k}^{-1} z^{-\frac{1}{2}} \eta\left(\frac{1}{k}\left(\left[-h\right]_k + \frac{i}{z}\right)\right).$$

Moreover, we require the following properties of $\vartheta(w;\tau)$ from [4, Lemma 2.1 (1), (2), (3)], [19, Proposition 1.3 (7), (8)], and [9].

Lemma 2.2. *Suppose that* $w \in \mathbb{C}$ *and* $\tau \in \mathbb{H}$*.*

(1) *Suppose that* $z \in \mathbb{C}$ *with* $\mathrm{Re}(z) > 0$*,* $k \in \mathbb{N}$*, and* $h \in \mathbb{Z}$ *with* $\gcd(h,k) = 1$*. Then we have*

$$\vartheta\left(w; \frac{1}{k}(h+iz)\right) = e^{\frac{\pi i}{4k}(h - [-h]_k)} e^{-\frac{3\pi i}{4}} \omega_{h,k}^{-3} \sqrt{\frac{i}{z}} e^{-\frac{\pi k w^2}{z}} \vartheta\left(\frac{iw}{z}; \frac{1}{k}\left(\left[-h\right]_k + \frac{i}{z}\right)\right).$$

(2) *Suppose that* $\lambda, \mu \in \mathbb{Z}$*. Then, with* $\zeta := e^{2\pi i w}$*, we have*

$$\vartheta(w + \lambda\tau + \mu; \tau) = (-1)^{\lambda+\mu} q^{-\frac{\lambda^2}{2}} \zeta^{-\lambda} \vartheta(w;\tau).$$

(3) *We have*

$$\vartheta(w;\tau) = -iq^{\frac{1}{8}} e^{-\pi i w} \prod_{n \ge 1} \left(1 - q^n\right)\left(1 - e^{2\pi i w} q^{n-1}\right)\left(1 - e^{-2\pi i w} q^n\right).$$

(4) *We have*

$$\vartheta\left(\frac{w}{\tau}; -\frac{1}{\tau}\right) = -i\sqrt{-i\tau} e^{\frac{\pi i w^2}{\tau}} \vartheta\left(w;\tau\right).$$

(5) *For* $\lambda \in \mathbb{Q}$*,* $q^{\frac{\lambda^2}{2}} \vartheta(\lambda\tau;\tau)$ *is a modular form of weight* $\frac{1}{2}$ *(on some congruence subgroup).*

Moreover, using [4, Lemma 2.1 (3)], Lemma 2.2 (3), (4), we have the following lemma.

Lemma 2.3. *Suppose that* $\tau \in \mathbb{H}$ *and* $a, b \in \mathbb{R}$*.*

(1) *If* $0 < a < 1$*, then we have, as* $\tau \to i\infty$*,*

$$\vartheta(a\tau + b; \tau) = -ie^{-\pi i b} q^{\frac{1}{8} - \frac{a}{2}} \left(1 + O\left(q^{\min\{a, 1-a\}}\right)\right).$$

[1]For the explicit shape of $\omega_{h,k}$, see [1, (5.2.4)]. We however do not require it in this paper.

(2) *For $b \notin \mathbb{Z}$, we have, as $\tau \to i\infty$,*

$$\vartheta(b;\tau) \sim -2\sin(\pi b)\, e^{\frac{\pi i \tau}{4}}.$$

(3) *If $0 < b < 1$, then we have, as $z \to 0$,*

$$\vartheta(aiz+b;iz) \sim -\frac{e^{-2\pi iab+\pi ia} e^{\left(-b^2-\frac{1}{4}+b\right)\frac{\pi}{z}}}{\sqrt{z}}.$$

(4) *For $a \notin \mathbb{Z}$, we have, as $z \to 0$,*

$$\vartheta(aiz;iz) \sim -\frac{2i\sin(\pi a)}{\sqrt{z}} e^{-\frac{\pi}{4z}}.$$

We also require the following asymptotics of the I-Bessel function.

Lemma 2.4. *We have, as $x \to \infty$,*

$$I_1(x) \sim \frac{e^x}{\sqrt{2\pi x}}.$$

2.3. **A Tauberian theorem.** We require a modified version of Ingham's Tauberian Theorem [12], stated in [5, Theorem 1.1]. For $\Delta \geq 0$, define

$$R_\Delta := \{x+iy : x, y \in \mathbb{R},\, x > 0,\, |y| \leq \Delta x\}.$$

Proposition 2.5. *Let $B(q) = \sum_{n\geq 0} b(n) q^n$ be a power series whose radius of convergence is at least one and assume that the $b(n)$ are non-negative and weakly increasing. Also suppose that ν, β, $\gamma \in \mathbb{R}$ with $\gamma > 0$ exist such that*[2]

$$B\left(e^{-t}\right) \sim \nu t^\beta e^{\frac{\gamma}{t}} \quad \text{as } t \to 0^+, \qquad B\left(e^{-z}\right) \ll |z|^\beta e^{\frac{\gamma}{|z|}} \quad \text{as } z \to 0, \tag{2.1}$$

with the latter condition holding in each region[3] *R_Δ for $\Delta \geq 0$. Then we have*

$$b(n) \sim \frac{\nu \gamma^{\frac{\beta}{2}+\frac{1}{4}}}{2\sqrt{\pi} n^{\frac{\beta}{2}+\frac{3}{4}}} e^{2\sqrt{\gamma n}} \qquad \text{as } n \to \infty.$$

Moreover, we require the asymptotic behavior of $(q)_\infty$ as $q \to 1$.

Lemma 2.6. *We have, with $q = e^{-z}$,*

$$(q)_\infty \sim \sqrt{\frac{2\pi}{z}} e^{-\frac{\pi^2}{6z}} \quad (\text{as } z \to 0).$$

3. Proof of Theorem 1.1 and Corollary 1.2

In this section, we consider the case that m is odd. For $m \in \mathbb{N}_{>3}$ and $\ell \in \mathbb{N}$, we study the behavior of $G_m^\ell(\tau)$, where

$$G_m(\tau) := \frac{\vartheta(\tau; m\tau)}{\vartheta(2\tau; m\tau)} = q^{\frac{1}{2}} F_m(q),$$

as $\tau \to \frac{h}{k} \in \mathbb{Q}$. For $a \in \mathbb{R}$, we write $a = \lfloor a \rfloor + \{a\}$. We have the following lemma (for general m). For $m \in \mathbb{N}_{>3}$, $\ell, k \in \mathbb{N}$, $d := \gcd(m,k)$ with $\mu := \frac{m}{d}$, and $\kappa := \frac{k}{d}$, define

$$\mathcal{A}_{m,\ell}(h,k) := \frac{\pi\ell}{\mu\kappa}\left(\left\{\frac{\mu h}{m}\right\} - \left\{\frac{\mu h}{m}\right\}^2 - \left\{\frac{2\mu h}{m}\right\} + \left\{\frac{2\mu h}{m}\right\}^2\right).$$

[2]Throughout we use the principal branch to define the logarithm and, consequently, the square root.

[3]Throughout, we consider $z \to 0$ in R_Δ with $\Delta \geq 0$.

Lemma 3.1. *Let $m \in \mathbb{N}_{>3}$ and $\ell, k \in \mathbb{N}$. For $0 \le h < k$ with $\gcd(h,k)=1$, we have, as $z \to 0$,*

$$G_m^{\ell}\left(\frac{h}{k}+\frac{iz}{k}\right) = e^{\frac{\mathcal{A}_{m,\ell}(h,k)}{z}+O_{h,k}(1)}.$$

Proof. It is enough to prove the claim for $\ell = 1$. Indeed, the general case follows by taking the ℓ-th power. Define

$$g_m(\tau) := G_m\left(\frac{\tau}{m}\right) = \frac{\vartheta\left(\frac{\tau}{m};\tau\right)}{\vartheta\left(\frac{2\tau}{m};\tau\right)}. \tag{3.1}$$

Then

$$\mathrm{Log}\left(g_m\left(\frac{1}{k}(h+iz)\right)\right) \equiv \sum_{j\in\{0,1\}} (-1)^j \mathrm{Log}\left(\vartheta\left(\frac{j+1}{mk}(h+iz); \frac{1}{k}(h+iz)\right)\right), \tag{3.2}$$

where, here and throughout, we use $\equiv$ to denote that the identity holds modulo $2\pi i\mathbb{Z}$. Now by Lemma 2.2 (1) with $w = \frac{j+1}{mk}(h+iz)$ $(j \in \{0,1\})$, we have, as $z \to 0$,

$$\begin{aligned}
&\mathrm{Log}\left(\vartheta\left(\frac{j+1}{mk}(h+iz); \frac{1}{k}(h+iz)\right)\right)\\
&\quad\equiv -\frac{\mathrm{Log}(z)}{2} - \frac{\pi(j+1)^2h^2}{m^2kz} + \mathrm{Log}\left(\vartheta\left(\frac{i(j+1)h}{mkz} - \frac{j+1}{mk}; \frac{1}{k}\left([-h]_k + \frac{i}{z}\right)\right)\right) + O_{h,k}(1).
\end{aligned}$$

Plugging this into (3.2), we obtain, as $z \to 0$,

$$\begin{aligned}
&\mathrm{Log}\left(g_m\left(\frac{1}{k}(h+iz)\right)\right)\\
&\qquad\equiv \frac{3\pi h^2}{m^2kz} + \sum_{j\in\{0,1\}} (-1)^j \mathrm{Log}\left(\vartheta\left(\frac{i(j+1)h}{mkz} - \frac{j+1}{mk}; \frac{1}{k}\left([-h]_k + \frac{i}{z}\right)\right)\right) + O_{h,k}(1)\\
&\qquad= \frac{3\pi h^2}{m^2kz} + \sum_{j\in\{0,1\}} (-1)^j \mathrm{Log}\left(\vartheta\left((j+1)\left(\frac{h}{m}\tau_1 + \alpha_1\right); \tau_1\right)\right) + O_{h,k}(1),
\end{aligned} \tag{3.3}$$

where

$$\alpha_1 := -\frac{h[-h]_k + 1}{mk} \quad \text{and} \quad \tau_1 := \frac{1}{k}\left([-h]_k + \frac{i}{z}\right).$$

By Lemma 2.2 (2) with $\lambda = \lfloor a \rfloor,\ \mu = 0,\ w = \{a\}\tau + b$, we have

$$\vartheta(a\tau+b;\tau) = (-1)^{\lfloor a\rfloor} e^{-2\pi i \lfloor a\rfloor b} q^{-\frac{\lfloor a\rfloor^2}{2} - \lfloor a\rfloor\{a\}} \vartheta(\{a\}\tau + b;\tau).$$

Applying this with $a = \frac{h}{m}$, $b = \alpha_1$, and $\tau = \tau_1$, we obtain,

$$\begin{aligned}
&\mathrm{Log}\left(\vartheta\left(\left(\left\lfloor\frac{h}{m}\right\rfloor + \left\{\frac{h}{m}\right\}\right)\tau_1 + \alpha_1; \tau_1\right)\right)\\
&\qquad\equiv -\pi i \left\lfloor\frac{h}{m}\right\rfloor^2 \tau_1 - 2\pi i \left\lfloor\frac{h}{m}\right\rfloor \left\{\frac{h}{m}\right\}\tau_1 + \mathrm{Log}\left(\vartheta\left(\left\{\frac{h}{m}\right\}\tau_1 + \alpha_1; \tau_1\right)\right) + O_{h,k}(1).
\end{aligned} \tag{3.4}$$

Now, by Lemma 2.3 (1), (2) with $a = \{\frac{h}{m}\}$, $b = \alpha_1$, and $\tau = \tau_1$, we have,

$$\mathrm{Log}\left(\vartheta\left(\left\{\frac{h}{m}\right\}\tau_1 + \alpha_1; \tau_1\right)\right) \equiv \frac{\pi i}{4}\tau_1 - \pi i\left\{\frac{h}{m}\right\}\tau_1 + O_{h,k}(1) \quad (\text{as } z \to 0).$$

Plugging this into (3.4), we obtain, as $z \to 0$,

$$\mathrm{Log}\left(\vartheta\left(\left(\left\lfloor\frac{h}{m}\right\rfloor + \left\{\frac{h}{m}\right\}\right)\tau_1 + \alpha_1; \tau_1\right)\right)$$

$$\equiv \frac{\pi i}{4}\tau_1 + \pi\left(\frac{h^2}{m^2} + \left\{\frac{h}{m}\right\} - \left\{\frac{h}{m}\right\}^2\right)\frac{1}{kz} + O_{h,k}(1). \tag{3.5}$$

In the same way, we have, as $z \to 0$,

$$\operatorname{Log}\left(\vartheta\left(\left(\left\lfloor\frac{2h}{m}\right\rfloor + \left\{\frac{2h}{m}\right\}\right)\tau_1 + 2\alpha_1; \tau_1\right)\right)$$
$$\equiv \frac{\pi i}{4}\tau_1 + \pi\left(\frac{4h^2}{m^2} + \left\{\frac{2h}{m}\right\} - \left\{\frac{2h}{m}\right\}^2\right)\frac{1}{kz} + O_{h,k}(1).$$

Plugging this and (3.5) into (3.3), we obtain, as $z \to 0$,

$$\operatorname{Log}\left(g_m\left(\frac{1}{k}(h+iz)\right)\right) \equiv \frac{\pi}{kz}\left(\left\{\frac{h}{m}\right\} - \left\{\frac{h}{m}\right\}^2 - \left\{\frac{2h}{m}\right\} + \left\{\frac{2h}{m}\right\}^2\right) + O_{h,k}(1).$$

Exponentiating and using (3.1) with $\tau \mapsto m\tau$, $m = d\mu$, and $k = d\kappa$, we conclude the proof. □

Next, we maximize $|G_m^\ell(\frac{1}{k}(h+iz))|$.

Lemma 3.2. *For $m \in \mathbb{N}_{>3}$ odd and $\ell \in \mathbb{N}$, $|G_m^\ell(\frac{1}{k}(h+iz))|$ is maximal, as $z \to 0$, for $\frac{h}{k} \in \{\frac{m-1}{2m}, \frac{m+1}{2m}\}$.*

Proof. Again we may assume $\ell = 1$. By Lemma 3.1, as $z \to 0$, $|G_m(\frac{1}{k}(h+iz))|$ is maximal for the maximal value of

$$\begin{cases} \frac{1}{\mu\kappa}\left(3\left\{\frac{\mu h}{m}\right\}^2 - \left\{\frac{\mu h}{m}\right\}\right) & \text{if} \quad 0 \le \left\{\frac{\mu h}{m}\right\} < \frac{1}{2}, \\ \frac{1}{\mu\kappa}\left(3\left\{\frac{\mu h}{m}\right\}^2 - 5\left\{\frac{\mu h}{m}\right\} + 2\right) & \text{if} \quad \frac{1}{2} \le \left\{\frac{\mu h}{m}\right\} < 1. \end{cases} \tag{3.6}$$

Let $d \mid m$. If $d = 1$, then $\mu = m$, $\kappa = k$, and $\{\frac{\mu h}{m}\} = \{h\} = 0$. So, by Lemma 3.1, $G_m(\frac{1}{k}(h+iz)) = O_{h,k}(1)$, as $z \to 0$. We next assume $d > 1$. Plugging in $\mu = \frac{m}{d}$ and $\kappa = \frac{k}{d}$, by (3.6), we need to maximize (note that d is odd since m is odd)

$$\begin{cases} \frac{d^2}{mk}\left(3\left\{\frac{h}{d}\right\}^2 - \left\{\frac{h}{d}\right\}\right) & \text{if} \quad 0 \le \left\{\frac{h}{d}\right\} < \frac{1}{2}, \\ \frac{d^2}{mk}\left(3\left\{\frac{h}{d}\right\}^2 - 5\left\{\frac{h}{d}\right\} + 2\right) & \text{if} \quad \frac{1}{2} \le \left\{\frac{h}{d}\right\} < 1, \end{cases}$$
$$= \begin{cases} \frac{d^2}{mk}\left(\frac{3j^2}{d^2} - \frac{j}{d}\right) & \text{if} \quad h \in d\mathbb{N}_0 + j,\ 0 \le j \le \left\lfloor\frac{d-1}{2}\right\rfloor, \\ \frac{d^2}{mk}\left(\frac{3j^2}{d^2} - \frac{5j}{d} + 2\right) & \text{if} \quad h \in d\mathbb{N}_0 + j,\ \left\lfloor\frac{d+1}{2}\right\rfloor \le j \le d-1. \end{cases}$$

Since m is odd and $d \mid m$, making the change of variables $j \mapsto d - j$ in the second case, we thus need to maximize

$$\begin{cases} \frac{d^2}{mk}\left(\frac{3j^2}{d^2} - \frac{j}{d}\right) & \text{if} \quad h \in d\mathbb{N}_0 + j,\ 0 \le j \le \frac{d-1}{2}, \\ \frac{d^2}{mk}\left(\frac{3j^2}{d^2} - \frac{j}{d}\right) & \text{if} \quad h \in d\mathbb{N} - j,\ 1 \le j \le \frac{d-1}{2}. \end{cases} \tag{3.7}$$

Observe that for $d = 3$ these expressions vanish. So for the remainder of the proof, we assume $d > 3$. For m and d fixed the expressions in (3.7) are maximized for $j = \frac{d-1}{2}$. The corresponding contribution is $\frac{(d-1)(d-3)}{4mk}$. Moreover, one sees that this is maximal for k minimal with $\gcd(m,k) = d$, which occurs for $k = d$. The expression we need to maximize in this case is $\frac{(d-1)(d-3)}{4md}$. As $d \mid m$, this is maximized for $d = m$. The corresponding values of h are $\frac{m-1}{2}$ and $\frac{m+1}{2}$ which are clearly coprime to $k = m$. □

Lemma 3.3. *For $m \in \mathbb{N}_{>3}$ odd, $\ell \in \mathbb{N}$, and $h \in \{\frac{m-1}{2}, \frac{m+1}{2}\}$, we have, as $z \to 0$,*

$$F_m^\ell\left(e^{\frac{2\pi i}{m}(h+iz)}\right) \sim C_{m,h}^\ell e^{\left(\frac{1}{4}-\frac{1}{m}+\frac{3}{4m^2}\right)\frac{\pi\ell}{z}}$$

with

$$C_{m,h} := \begin{cases} -ie^{\frac{5\pi i}{2m}-\frac{3\pi i}{m^2}} & \text{if } h = \frac{m-1}{2}, \\ ie^{-\frac{5\pi i}{2m}+\frac{3\pi i}{m^2}} & \text{if } h = \frac{m+1}{2}. \end{cases}$$

Proof. Note that it is enough to prove the case $\ell = 1$. By Lemma 2.2 (3), for $m \in \mathbb{N}_{>3}$ odd, we have, as $z \to 0$,

$$F_m\left(e^{\frac{2\pi i}{m}\left(\frac{m-1}{2}+iz\right)}\right) \sim -ie^{\frac{\pi i}{2m}} \frac{\vartheta\left(\frac{1}{2}-\frac{1}{2m}+\frac{iz}{m}; iz\right)}{\vartheta\left(1-\frac{1}{m}+\frac{2iz}{m}; iz\right)}.$$

Moreover, by definition, $c_m^{[1]}(n) \in \mathbb{R}$. So, $F_m(\overline{q}) = \overline{F_m(q)}$, and thus we only need to prove the claim for $\frac{m-1}{2}$. Lemma 2.3 (3) finishes the proof. □

Proof of Theorem 1.1. By Lemma 2.2 (5), $q^{-\frac{3\ell}{2m}+\frac{\ell}{2}} F_m^\ell(q) = q^{-\frac{3\ell}{2m}} G_m^\ell(\tau)$ is a modular function. By Lemma 3.2 its maximal growth occurs at the cusps $\frac{m-1}{2m}$ and $\frac{m+1}{2m}$. Using Lemma 2.4, Lemma 3.3, and (1.3), Theorem 2.1 now gives, as $n \to \infty$,

$$c_m^{[\ell]}(n) \sim \left(\frac{\left(1-\frac{1}{m}\right)\left(1-\frac{3}{m}\right)\ell}{2m}\right)^{\frac{1}{4}} \frac{(-1)^n \cos\left(\pi\left(\phi_m\ell - \frac{n}{m}\right)\right) e^{\pi\sqrt{\frac{2\left(1-\frac{1}{m}\right)\left(1-\frac{3}{m}\right)\ell}{m}n}}}{n^{\frac{3}{4}}}.$$

Here we require that the right-hand side does not vanish, which we now prove. Precisely, we need to show that, for $n \in \mathbb{N}$ and $\ell \in \mathbb{N}$,

$$\cos\left(\pi\left(\phi_m\ell - \frac{n}{m}\right)\right) \neq 0.$$

For this, it suffices to prove that $\phi_m\ell - \frac{n}{m} \notin \mathbb{Z} + \frac{1}{2}$. Suppose for contradiction that, for some $j \in \mathbb{Z}$,

$$\phi_m\ell - \frac{n}{m} = j + \frac{1}{2} \Rightarrow (m-2)(m-3)\ell - 2mn = 2m^2 j + m^2,$$

which cannot hold because the left-hand side is even whereas the right-hand side is odd, as m is odd. This concludes the proof. □

Proof of Corollary 1.2. By Theorem 1.1, for $n \gg 1$ with $n \equiv r \pmod{2m}$,

$$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right) = (-1)^r \operatorname{sgn}\left(\cos\left(\pi\left(\phi_m\ell - \frac{r}{m}\right)\right)\right). \tag{3.8}$$

Thus, it suffices to determine $\operatorname{sgn}(\cos(\pi(\phi_m\ell - \frac{r}{m})))$. We only consider the case ℓ even, the case ℓ odd is shown analogously. Using (1.3), we write

$$\pi\left(\phi_m\ell - \frac{r}{m}\right) = \pi\left(\frac{\ell}{2} + \frac{3\ell}{m^2} - \frac{R}{m}\right), \tag{3.9}$$

where

$$R := R_{r,\ell} := r + \frac{5\ell}{2}. \tag{3.10}$$

Using (3.9), we obtain

$$\cos\left(\pi\left(\phi_m\ell - \frac{r}{m}\right)\right) = (-1)^{\frac{\ell}{2}} \cos\left(\pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right)\right). \tag{3.11}$$

Now note that by (3.10), we have

$$5 \leq R < \frac{5m}{2}.$$

Next, we split the range of R. For $5 \leq R \leq \frac{m-1}{2}$, we have

$$0 < \pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right) < \frac{\pi}{2}.$$

Thus

$$\operatorname{sgn}\left(\cos\left(\pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right)\right)\right) = 1 \quad \text{for} \quad 5 \leq R \leq \frac{m-1}{2}. \tag{3.12}$$

Next, for $\frac{m+1}{2} \leq R \leq \frac{3m-1}{2}$, we have

$$\frac{\pi}{2} < \pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right) < \frac{3\pi}{2}.$$

Thus, we obtain

$$\operatorname{sgn}\left(\cos\left(\pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right)\right)\right) = -1 \quad \text{for} \quad \frac{m+1}{2} \leq R \leq \frac{3m-1}{2}. \tag{3.13}$$

Finally, for the remaining range of R, i.e., $\frac{3m+1}{2} \leq R < \frac{5m}{2}$, we have

$$\frac{3\pi}{2} < \pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right) < \frac{5\pi}{2}.$$

Thus, we obtain

$$\operatorname{sgn}\left(\cos\left(\pi\left(\frac{R}{m} - \frac{3\ell}{m^2}\right)\right)\right) = 1 \quad \text{for} \quad \frac{3m+1}{2} \leq R < \frac{5m}{2}.$$

Combining this, (3.12), (3.13), (3.11), (3.10), and (3.8), we obtain the claim. □

4. Proof of Theorem 1.3 and Corollary 1.4

In this section, we consider the case that m is even. We first determine the asymptotics of $F_m^\ell(-e^{-z})$ as $z \to 0$. Note that, by the fact m even and (1.2), we have

$$F_m^\ell(-q) = \frac{\left(-q, -q^{m-1}; q^m\right)_\infty^\ell}{\left(q^2, q^{m-2}; q^m\right)_\infty^\ell} = \sum_{n \geq 0} (-1)^n c_m^{[\ell]}(n) q^n. \tag{4.1}$$

We set

$$d_m^{[\ell]}(n) := (-1)^n c_m^{[\ell]}(n). \tag{4.2}$$

Lemma 4.1. *For $n \in \mathbb{N}_0$, $d_m^{[\ell]}(n) \geq 0$ and $\{d_m^{[\ell]}(n)\}_{n \geq 0}$ is weakly increasing.*

Proof. From (4.1), it directly follows that $d_m^{[\ell]}(n) \geq 0$ for $n \in \mathbb{N}_0$.

To prove that $\{d_m^{[\ell]}(n)\}_{n \geq 0}$ is weakly increasing, it suffices to show that $(1-q)F_m^\ell(-q)$ has non-negative coefficients. By (4.1), we have

$$(1-q)\, F_m^\ell(-q) = \frac{\left(-q^{m+1}, -q^{m-1}; q^m\right)_\infty^\ell}{(1-q)^{\ell-1} \left(q^{m+2}, q^{m-2}; q^m\right)_\infty^\ell}$$

which has non-negative coefficients. This concludes the proof. □

Next, we determine the asymptotics of $F_m^\ell(-e^{-z})$, as $z \to 0$.

Lemma 4.2. *We have, as $z \to 0$,*

$$F_m^{\ell}\left(-e^{-z}\right) \sim \frac{e^{\frac{\pi^2 \ell}{2mz}}}{2^{\ell}\sin^{\ell}\left(\frac{2\pi}{m}\right)}.$$

Proof. Note that it is enough to prove the case $\ell = 1$. By (4.1) and Lemma 2.2 (3), we have

$$F_m(-q) = -\frac{ie^{-\pi i\tau}\left(q^m;q^m\right)_{\infty}^2 \vartheta\left(2\tau;2m\tau\right)}{\left(q^{2m};q^{2m}\right)_{\infty}\vartheta\left(\tau;m\tau\right)\vartheta\left(2\tau;m\tau\right)}.$$

Using this with $\tau = \frac{iz}{2\pi}$, Lemma 2.6 and Lemma 2.3 (4) imply the claim. □

Proof of Theorem 1.3. By Lemma 4.1, $d_m^{[\ell]}(n)$ satisfies the monotonicity and non-negativity required to apply Proposition 2.5. Moreover, Lemma 4.2 shows that $F_m^{\ell}(-e^{-z})$ satisfies (2.1) with $\nu = \frac{1}{2^{\ell}\sin^{\ell}(\frac{2\pi}{m})}$, $\beta = 0$, and $\gamma = \frac{\pi^2\ell}{2m}$. By Proposition 2.5, we have, as $n \to \infty$,

$$d_m^{[\ell]}(n) \sim \frac{\ell^{\frac{1}{4}}}{2^{\ell+\frac{5}{4}}m^{\frac{1}{4}}\sin^{\ell}\left(\frac{2\pi}{m}\right)}\frac{e^{\pi\sqrt{\frac{2\ell}{m}n}}}{n^{\frac{3}{4}}}.$$

By (4.2), we conclude the proof. □

Proof of Corollary 1.4. The claim follows from Lemma 4.1 and (4.2). □

5. Proof of Theorem 1.5 and Corollary 1.6

For $m \in \mathbb{N}_{>3}$ and $\lambda \in \mathbb{N}$, we study the main growth of $\mathcal{G}_m^{\lambda}(\tau)$, where

$$\mathcal{G}_m(\tau) := \frac{\vartheta\left(2\tau;m\tau\right)}{\vartheta\left(\tau;m\tau\right)} = q^{-\frac{1}{2}}\mathcal{F}_m(q),$$

as $\tau \to \frac{h}{k} \in \mathbb{Q}$. Noting that $\mathcal{G}_m = G_m^{-1}$, Lemma 3.1 implies the following:

Lemma 5.1. *Let $m \in \mathbb{N}_{>3}$ and $\lambda, k \in \mathbb{N}$. For $0 \le h < k$ with $\gcd(h,k) = 1$, $d := \gcd(m,k)$ with $\mu := \frac{m}{d}$ and $\kappa := \frac{k}{d}$, we have,*

$$\mathcal{G}_m^{\lambda}\left(\frac{h}{k} + \frac{iz}{k}\right) = e^{-\frac{\mathcal{A}_{m,\lambda}(h,k)}{z} + O_{h,k}(1)} \quad (\text{as } z \to 0).$$

Next, we maximize $|\mathcal{G}_m^{\lambda}(\frac{1}{k}(h+iz))|$.

Lemma 5.2. *For $m \in \mathbb{N}_{>3}$ with $m \equiv \pm 1 \pmod 6$ and $\lambda \in \mathbb{N}$, $|\mathcal{G}_m^{\lambda}(\frac{1}{k}(h+iz))|$ is maximal, as $z \to 0$, for $\frac{h}{k} \in \{\frac{m\mp 1}{6m}, \frac{5m\pm 1}{6m}\}$.*

Proof. By taking the negative of (3.6) we see that, as $z \to 0$, $|\mathcal{G}_m^{\lambda}(\frac{1}{k}(h+iz))|$ is maximal for the maximal value of

$$\begin{cases} \frac{1}{\mu\kappa}\left(-3\left\{\frac{\mu h}{m}\right\}^2 + \left\{\frac{\mu h}{m}\right\}\right) & \text{if} \quad 0 \le \left\{\frac{\mu h}{m}\right\} < \frac{1}{2}, \\ \frac{1}{\mu\kappa}\left(-3\left\{\frac{\mu h}{m}\right\}^2 + 5\left\{\frac{\mu h}{m}\right\} - 2\right) & \text{if} \quad \frac{1}{2} \le \left\{\frac{\mu h}{m}\right\} < 1. \end{cases} \tag{5.1}$$

Let $d \mid m$. If $d = 1$, then $\mu = m$, $\kappa = k$, and $\{\frac{\mu h}{m}\} = \{h\} = 0$. So, by Lemma 5.1, $\mathcal{G}_m^{\lambda}(\frac{1}{k}(h+iz)) = O_{h,k}(1)$ as $z \to 0$. Next, we assume $d > 1$. Plugging in $\mu = \frac{m}{d}$ and $\kappa = \frac{k}{d}$, by (5.1), making the change of variables $j \mapsto d - j$ in the second case, thus we need to maximize

$$\begin{cases} \frac{d^2}{mk}\left(-\frac{3j^2}{d^2} + \frac{j}{d}\right) & \text{if} \quad h \in d\mathbb{N}_0 + j,\ 0 \le j \le \frac{d-1}{2}, \\ \frac{d^2}{mk}\left(-\frac{3j^2}{d^2} + \frac{j}{d}\right) & \text{if} \quad h \in d\mathbb{N} - j,\ 1 \le j \le \frac{d-1}{2}. \end{cases} \tag{5.2}$$

We note that for m and d fixed these expressions are maximized if $|6j-d|$ is minimal. Now $|6j-d|$ is minimal for $j=\frac{d\mp 1}{6}$. In this case, (5.2) is maximal for $k=d$. The expression we need to maximize in this case is $\frac{d-\frac{1}{d}}{12m}$. This is maximized for $d=m$. The maximum value is $\frac{1}{12}(1-\frac{1}{m^2})$, which is positive for $m\in\mathbb{N}_{>3}$. Since $j=\frac{m\mp 1}{6}$, we have $m=6j\pm 1$ and hence

$$\gcd(m,j)=\gcd(6j\pm 1,j)=1.$$

Thus, for $h\in m\mathbb{N}_0+j$ (for the plus sign) and $k=m$ (resp. $h\in m\mathbb{N}-j$ and $k=m$ for the minus sign), we have $\gcd(h,k)=1$. Taking the representative $0\le h<m$ gives $h=j=\frac{m\mp 1}{6}$ and $h=m-j=\frac{5m\pm 1}{6}$. □

Next, we determine the growth of $\mathcal{F}_m^\lambda(e^{\frac{2\pi i}{m}(h+iz)})$ as $z\to 0$.

Lemma 5.3. *For $m\in\mathbb{N}_{>3}$ with $m\equiv\pm 1 \pmod 6$, $\lambda\in\mathbb{N}$, and $h\in\{\frac{m\mp 1}{6},\frac{5m\pm 1}{6}\}$, we have,*

$$\mathcal{F}_m^\lambda\left(e^{\frac{2\pi i}{m}(h+iz)}\right)\sim \mathcal{C}_{m,h}^\lambda e^{\left(\frac{1}{12}-\frac{1}{12m^2}\right)\frac{\pi\lambda}{z}}\quad (as\ z\to 0),$$

where

$$\mathcal{C}_{m,h}:=\begin{cases} e^{\frac{\pi i}{6}\mp\frac{\pi i}{6m}\pm\frac{\pi i}{m^2}} & \text{if } h=\frac{m\mp 1}{6},\\ e^{-\frac{\pi i}{6}\pm\frac{\pi i}{6m}\mp\frac{\pi i}{m^2}} & \text{if } h=\frac{5m\pm 1}{6}.\end{cases}$$

Proof. Again it is enough to consider the case $\lambda=1$. Moreover, by definition, $\gamma_m^{[1]}(n)\in\mathbb{R}$. So, $\mathcal{F}_m(\overline{q})=\overline{\mathcal{F}_m(q)}$ and thus we only need to prove the claim for $h=\frac{m\mp 1}{6}$. Using Lemma 2.2 (3), for $h=\frac{m\mp 1}{6}$, we have, as $z\to 0$,

$$\mathcal{F}_m\left(e^{\frac{2\pi i}{m}\left(\frac{m\mp 1}{6}+iz\right)}\right)\sim e^{\frac{\pi i}{6}}e^{\mp\frac{\pi i}{6m}}\frac{\vartheta\left(\frac{1}{3}\mp\frac{1}{3m}+\frac{2iz}{m};iz\right)}{\vartheta\left(\frac{1}{6}\mp\frac{1}{6m}+\frac{iz}{m};iz\right)}.$$

Now, Lemma 2.3 (3) finishes the proof. □

Proof of Theorem 1.5. By Lemma 2.2 (5), $q^{\frac{3\lambda}{2m}-\frac{\lambda}{2}}\mathcal{F}_m^\lambda(q)$ is a modular function. By Lemma 5.2 its maximal growth occurs at the cusps $\frac{m\mp 1}{6m}$ and $\frac{5m\pm 1}{6m}$. Using Lemma 5.3 and Lemma 2.4, Theorem 2.1 now gives, as $n\to\infty$,

$$\gamma_m^{[\lambda]}(n)\sim\left(\frac{\left(1-\frac{1}{m}\right)\left(1+\frac{1}{m}\right)\lambda}{6m}\right)^{\frac{1}{4}}\frac{T_m^{[\lambda]}(n)e^{\pi\sqrt{\frac{2\left(1-\frac{1}{m}\right)\left(1+\frac{1}{m}\right)\lambda}{3m}n}}}{n^{\frac{3}{4}}}.$$

Here we require that the right-hand side does not vanish, which we now prove. Precisely, for $n,\lambda\in\mathbb{N}$ and $m\in\mathbb{N}_{>3}$ with $\gcd(m,6)=1$, we need to show

$$T_m^{[\lambda]}(n)\neq 0.$$

The proof is analogous to the argument given in Theorem 1.1. □

Proof of Corollary 1.6. By Theorem 1.5, for $n\gg 1$,

$$\operatorname{sgn}\left(\gamma_m^{[\lambda]}(n)\right)=\operatorname{sgn}\left(T_m^{[\lambda]}(n)\right).$$

Since $n\equiv 3r+2mj \pmod{6m}$, we obtain, using that $m\equiv\pm 1 \pmod 6$,

$$T_m^{[\lambda]}(n)=(-1)^r\cos\left(\pi\left(\frac{\lambda}{6}\pm\frac{1}{m}\left(r-\frac{\lambda}{6}+\frac{\lambda}{m}\right)\right)\right). \tag{5.3}$$

We write $\lambda=6s+a$ with $0\le a\le 5$. Since $m\equiv\pm 1 \pmod 6$, we have

$$R=R_a:=a\frac{\pm m-1}{6}+r-s\in\mathbb{Z}.$$

Noting that

$$\frac{\lambda}{6} \pm \frac{1}{m}\left(r - \frac{\lambda}{6} + \frac{\lambda}{m}\right) = s \pm \frac{1}{m}\left(R + \frac{\lambda}{m}\right),$$

and using (5.3), we have,

$$T_m^{[\lambda]}(n) = (-1)^{r+s} \cos\left(\pi\left(\frac{R}{m} + \frac{\lambda}{m^2}\right)\right). \tag{5.4}$$

Since $0 < \lambda < \frac{m}{2}$ by assumption, we have

$$0 < \frac{\lambda}{m^2} < \frac{1}{2m}.$$

Thus, it is not hard to see that

$$\operatorname{sgn}\left(\cos\left(\pi\left(\frac{R}{m} + \frac{\lambda}{m^2}\right)\right)\right) = (-1)^{\chi_{R \pmod{2m} \in \left[\frac{m+1}{2}, \frac{3m-1}{2}\right]}}. \tag{5.5}$$

We now apply (5.4) and (5.5), distinguishing residue classes of λ modulo 6.

Case 1: $\lambda \equiv 0 \pmod 6$**.** Here $a = 0$ and $s = \frac{\lambda}{6}$. Thus $R = r - s$. We have

$$R \pmod{2m} \in \left[\frac{m+1}{2}, \frac{3m-1}{2}\right] \Leftrightarrow r \pmod{2m} \in \left[\frac{m+1}{2} + \frac{\lambda}{6}, \frac{3m-1}{2} + \frac{\lambda}{6}\right].$$

It is not hard to show that this interval lies in $[0, 2m-1]$. Thus we can skip the condition $r \pmod{2m}$. By (5.4), we conclude the claim in this case.

Case 2: $\lambda \equiv 1 \pmod 6$**.** As the proof of this case is analogous to the case $\lambda \equiv 0 \pmod 6$, we skip it here.

Case 3: $\lambda \equiv 2 \pmod 6$**.** Again the proof follows that of the case $\lambda \equiv 0 \pmod 6$ and we therefore skip it here.

Case 4: $\lambda \equiv 3 \pmod 6$**.** Here $a = 3$ and $s = \frac{\lambda - 3}{6}$. Then $R = \frac{\pm m - 1}{2} + r - s$. We have

$$R \pmod{2m} \in \left[\frac{m+1}{2}, \frac{3m-1}{2}\right] \Leftrightarrow r \pmod{2m} \in \left[\frac{m+1}{2} - \frac{\pm m - 1}{2} + s, \frac{3m-1}{2} - \frac{\pm m - 1}{2} + s\right].$$

We first assume that $m \equiv 1 \pmod 6$. Then our condition becomes

$$r \pmod{2m} \in \left[\frac{\lambda - 3}{6} + 1, m + \frac{\lambda - 3}{6}\right].$$

It is not hard to see that this again lies in $[0, 2m-1]$. The claim follows in this case as $(\frac{m}{3}) = 1$.

Next, assume that $m \equiv -1 \pmod 6$. Then our condition becomes

$$\begin{aligned} r \pmod{2m} &\in [m + s + 1, 2m + s] = [m + s + 1, 2m - 1] \cup [2m, 2m + s] \\ &\equiv [m + s + 1, 2m - 1] \cup [0, s] \pmod{2m}, \end{aligned}$$

which is the complement of $[s + 1, m + s]$ in $[0, 2m - 1]$. Thus

$$(-1)^{\chi_{r \in [m+s+1, 2m-1] \cup [0,s]}} = (-1)^{1 + \chi_{r \in [s+1, m+s]}}.$$

Noting that $(\frac{m}{3}) = -1$, the claim follows in this case.

Case 5: $\lambda \equiv 4 \pmod 6$**.** Here $a = 4$ and $s = \frac{\lambda - 4}{6}$. Then $R = \frac{2(\pm m - 1)}{3} + r - s$. Thus we have

$$R \pmod{2m} \in \left[\frac{m+1}{2}, \frac{3m-1}{2}\right]$$

$$\Leftrightarrow r \pmod{2m} \in \left[\frac{m+1}{2} - \frac{2\left(\pm m - 1\right)}{3} + s, \frac{3m-1}{2} - \frac{2\left(\pm m - 1\right)}{3} + s\right].$$

If $m \equiv 1 \pmod 6$, the above condition becomes

$$r \pmod{2m} \in \left[-\frac{m}{6} + \frac{\lambda-4}{6} + \frac{7}{6}, \frac{5m}{6} + \frac{\lambda-4}{6} + \frac{1}{6}\right].$$

Noting that the left endpoint is negative, we split the interval as

$$\left[-\frac{m}{6} + \frac{\lambda-4}{6} + \frac{7}{6}, -1\right] \cup \left[0, \frac{5m+1}{6} + \frac{\lambda-4}{6}\right]$$
$$\equiv \left[0, \frac{5m+1}{6} + \frac{\lambda-4}{6}\right] \cup \left[\frac{11m+7}{6} + \frac{\lambda-4}{6}, 2m-1\right] \pmod{2m}.$$

This is the complement of $[\frac{5m+7}{6} + \frac{\lambda-4}{6}, \frac{11m+1}{6} + \frac{\lambda-4}{6}]$ in $[0, 2m-1]$. The claim follows in this case.

We next consider the case $m \equiv -1 \pmod 6$. Then our condition becomes

$$r \pmod{2m} \in \left[\frac{7m+7}{6} + \frac{\lambda-4}{6}, \frac{13m+1}{6} + \frac{\lambda-4}{6}\right]. \tag{5.6}$$

Note that $\frac{7m+7}{6} + \frac{\lambda-4}{6} < 2m$ and $\frac{13m+1}{6} + \frac{\lambda-4}{6} > 2m$. Thus we write the interval in (5.6) as

$$\left[\frac{7m+7}{6} + \frac{\lambda-4}{6}, 2m-1\right] \cup \left[2m, \frac{13m+1}{6} + \frac{\lambda-4}{6}\right]$$
$$\equiv \left[0, \frac{m+1}{6} + \frac{\lambda-4}{6}\right] \cup \left[\frac{7m+7}{6} + \frac{\lambda-4}{6}, 2m-1\right] \pmod{2m}.$$

This is the complement of $[\frac{m+7}{6} + \frac{\lambda-4}{6}, \frac{7m+1}{6} + \frac{\lambda-4}{6}]$ in $[0, 2m-1]$, yielding the claim in this case.

Case 6: $\lambda \equiv 5 \pmod 6$**.** Here $a = 5$ and $s = \frac{\lambda-5}{6}$. Then $R = \frac{5(\pm m-1)}{6} + r - s$. We have

$$R \pmod{2m} \in \left[\frac{m+1}{2}, \frac{3m-1}{2}\right]$$
$$\Leftrightarrow r \pmod{2m} \in \left[\frac{m+1}{2} - \frac{5\left(\pm m - 1\right)}{6} + s, \frac{3m-1}{2} - \frac{5\left(\pm m - 1\right)}{6} + s\right].$$

We first assume that $m \equiv 1 \pmod 6$. Then our condition becomes

$$r \pmod{2m} \in \left[\frac{4-m}{3} + \frac{\lambda-5}{6}, \frac{2m+1}{3} + \frac{\lambda-5}{6}\right].$$

Noting that $\frac{4-m}{3} + \frac{\lambda-5}{6} < 0$, we rewrite the above interval as

$$\left[\frac{4-m}{3} + \frac{\lambda-5}{6}, -1\right] \cup \left[0, \frac{2m+1}{3} + \frac{\lambda-5}{6}\right]$$
$$\equiv \left[0, \frac{2m+1}{3} + \frac{\lambda-5}{6}\right] \cup \left[\frac{5m+4}{3} + \frac{\lambda-5}{6}, 2m-1\right] \pmod{2m}.$$

This is the complement of $[\frac{2m+4}{3} + \frac{\lambda-5}{6}, \frac{5m+1}{3} + \frac{\lambda-5}{6}]$ in $[0, 2m-1]$ and this claim follows in this case.

We next assume $m \equiv -1 \pmod 6$. In this case, our condition becomes

$$r \pmod{2m} \in \left[\frac{4m+4}{3} + \frac{\lambda-5}{6}, \frac{7m+1}{3} + \frac{\lambda-5}{6}\right].$$

Noting that $\frac{4m+4}{3}+\frac{\lambda-5}{6}<2m$ and $\frac{7m+1}{3}+\frac{\lambda-5}{6}>2m$, we rewrite the above interval as

$$\left[\frac{4m+4}{3}+\frac{\lambda-5}{6},2m-1\right]\cup\left[2m,\frac{7m+1}{3}+\frac{\lambda-5}{6}\right]$$
$$\equiv\left[0,\frac{m+1}{3}+\frac{\lambda-5}{6}\right]\cup\left[\frac{4m+4}{3}+\frac{\lambda-5}{6},2m-1\right]\pmod{2m}.$$

This is the complement of $[\frac{m+4}{3}+\frac{\lambda-5}{6},\frac{4m+1}{3}+\frac{\lambda-5}{6}]$ in $[0,2m-1]$ and the claim follows in this case. □

6. Open questions

The results of this paper suggest several directions for future research.

(1) It would be interesting to extend Theorems 1.1, 1.3, and 1.5, by considering theta quotients of the form (for $a,b,m,\ell\in\mathbb{N}$ with $a\neq b$)

$$\frac{\vartheta(a\tau;m\tau)^{\ell}}{\vartheta(b\tau;m\tau)^{\ell}}.$$

For which choices of a,b,m, and ℓ do the Fourier coefficients have an eventual periodic sign pattern? Can this sign pattern be determined explicitly? More generally, can one characterize the dominant cusps in terms of a,b, and m?

(2) In Theorem 1.5, we assume $\gcd(m,6)=1$. Can this assumption be removed? More precisely, how do the dominant cusps and the resulting sign pattern change if $2\mid m$ or $3\mid m$?

(3) Our results raise a more general question: under what conditions do the Fourier coefficients of a modular theta quotient have an eventually periodic sign pattern?

Appendix A. Numerics

In the following tables, we list the sign patterns of $c_m^{[\ell]}(n)$ and $\gamma_m^{[\ell]}(n)$ for $m\in\{5,7\}$, $1\leq\ell\leq 6$, and $70\leq n\leq m+69$.

(m,ℓ)	$(5,1)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$
$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right)$	$+-+--$	$+-+--$	$+-+-+$	$+-+-+$	$--+-+$
$\operatorname{sgn}\left(\gamma_m^{[\ell]}(n)\right)$	$++---$	$+++--$	$+++--$	$-++--$	$-+++-$

(m,ℓ)	$(5,6)$	$(7,1)$	$(7,2)$	$(7,3)$
$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right)$	$-++-+$	$+-+-++-$	$+-+-+-+$	$--+-+-+$
$\operatorname{sgn}\left(\gamma_m^{[\ell]}(n)\right)$	$--++-$	$+++---+$	$+++----$	$++++--$

(m,ℓ)	$(7,4)$	$(7,5)$	$(7,6)$
$\operatorname{sgn}\left(c_m^{[\ell]}(n)\right)$	$-+--+-+$	$-+-++-+$	$-+-+-++$
$\operatorname{sgn}\left(\gamma_m^{[\ell]}(n)\right)$	$-++++--$	$--+++--$	$--++++-$

Table 1. Sign patterns for $c_m^{[\ell]}(n)$ and $\gamma_m^{[\ell]}(n)$.

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University of Cologne, Department of Mathematics and Computer Science, Weyertal 86-90, 50931 Cologne, Germany

Email address: kbanerj1@uni-koeln.de

Email address: kbringma@math.uni-koeln.de

Email address: bheim@uni-koeln.de

The University of Hong Kong, Department of Mathematics, Pokfulam, Hong Kong

Email address: bkane@hku.hk